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A karyological study of the spiny mouse *Acomys* Geoffroy 1838 (Rodentia Muridae) along the Ethiopian Rift Valley

V.E. SOKOLOV¹, V.N. ORLOV¹, M.I. BASKEVICH¹, AFEWORK BEKELE² and ASSEFA
MEBRATE³

¹ *Institute of Evolutionary Animal Morphology and Ecology, Leninski 33, 117071
Moscow, Russia*

² *Biological Department, Addis Abeba University, Addis Abeba, Ethiopia*

³ *Science Commission, Addis Abeba, Ethiopia*

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Descriptions are provided of chromosomal sets and comparative analyses of C-banded chromosomes of the spiny mice from the genus *Acomys* Geoffroy 1838, which were collected from five localities in Ethiopia (Gamo-Gofa province: Konso area, Arba Minch; upper part of Awash valley: Koka area; middle part of Awash valley: Awash National Park and 70 km North from Awash National Park). Two karyotypic forms were discovered: one, as expected *A. cabirinus* Desmarest 1819 ($2n = 36$, NF = 68) and one new for the Ethiopian fauna 68-chromosomal *Acomys* sp. The distribution of these forms is discussed. Details of the C-banded chromosomes of *Acomys cabirinus* and *Acomys* sp. are given for the first time. It was shown that both these karyotypes possess mainly pericentromeric heterochromatin and have insignificant differences in their quantities. It is possible that heterochromatin variations did not make a contribution to the karyotypic evolution of *Acomys*. Apparently, chromosomal evolution in this genus was determined by Robertsonian or tandemic type translocations.

KEY WORDS: Rodentia, *Acomys*, systematics, karyotype, Ethiopia.

Introduction	227
Material and methods	228
Results	231
Discussion	233
Acknowledgements	234
References	234

INTRODUCTION

The modern classification of the spiny mice from the genus *Acomys* Geoffroy 1838 is very provisional. The number of species included in this genus varies: 26

(ALLEN 1939, ELLERMAN 1941), 18 (ANDERSON & JONES 1967), 9 (CORBET & HILL 1986), 5 (SETZER 1968, SOKOLOV 1977).

Based upon external morphology SETZER (1968) grouped the 35 African forms (not only species rank) into five species: *A. cahirinus* Desmarest 1819, *A. dimidiatus* Cretzmar 1826, *A. russatus* Wagner 1839, *A. spinosissimus* Peters 1852, *A. subspinosus* Waterhouse 1837, but the revision was unacceptable because it is not consistent with the available cytotaxonomy (WAHRMAN & ZAHAVI 1953; MATTHEY 1956, 1963, 1965, 1968) or dermal characteristics of various types. Based upon the peculiarities of the upper molar teeth the taxonomical revision of this genus assigned the African genus into three groups: complex *A. subspinosus*, *A. spinosissimus* and *A. russatus*; complex *cahirinus-dimidiatus* (species *A. percivali* Dollman 1911, *A. brockmani* Dollman 1911, *A. witherby* De Winton 1901, *A. ignitus* Dollman 1910 and *A. wilsoni* Thomas 1892) and subgenus *Peracomys* Petter & Roche 1981 [species *A. (P.) louisae* Thomas 1896 and *A. (P.) umbratus* Thomas 1923 (PETTER 1983)].

Preliminary karyological data (WAHRMAN & ZAHAVI 1953; MATTHEY 1956, 1963, 1965, 1968; MATTHEY & BACCAR 1967) was used to divide forms into three groups: those with low diploid numbers (*A. cahirinus*, *A. nesiotus* Bate 1903, *A. minous* Bate 1905, *A. seurati* Heim & Balsac 1937, *A. percivali*), those with high diploid numbers and ordinary heterochromosomes (*A. ignitus*, *A. russatus*, *A. wilsoni*) and a group of South African species with high diploid numbers and giant "triplicate" X-chromosomes (*A. subspinosus* and *A. selousi* de Winton 1896) (MATTHEY 1968). These and subsequent karyological studies (Table 1) demonstrated great chromosomal diversity between different forms of spiny mice and showed that *Acomys* is taxonomically a complicated group.

It is clear that there is no agreement between different approaches to the taxonomy of *Acomys*. As a result a new revision of this genus on the basis of all data, including chromosomal analysis is needed. The methods of cytotaxonomy can make a considerable contribution to the revision of this genus which is important from an epidemiological perspective [potential hosts of ecto- and endoparasites which may threaten human and animal health (AL-KHAHLI 1984, DIAB et al. 1985)], however, the systematic investigation of *Acomys* is also of interest in its own right.

The distribution of the genus *Acomys* is connected with arid, mainly rocky and only occasionally with sandy desert and semidesert regions in the South-West Palaearctica and in most of Africa.

The aim of the present work was to study chromosomal sets of the spiny mouse *Acomys* from five localities along the Ethiopian Rift Valley. These were ordinally grouped under two species *A. cahirinus* and *A. wilsoni* (YALDEN et al. 1976).

MATERIAL AND METHODS

The material used in the present investigation was obtained during a Soviet-Ethiopian biological field expedition seasons between 1988-1990.

The material consisted the mice from the genus *Acomys* collected from the following regions of Ethiopia along the Rift Valley:

Gamo-Gofa province: Konso area, 2 specimens; Arba Minch, 1 specimen.

Shoa province (upper part of Awash valley): Koka area, 18 specimens.

Shoa province (middle part of Awash valley): North bank of Awash river, (Awash National

Table 1.
Chromosomal data for the spiny mice from the genus *Acomys*.

Species and forms <i>Acomys</i>	2n	NF	X-morphology	Y-morphology	Banding	Country locality	Authors
1	2	3	4	5	6	7	8
<i>A. percivali</i> = <i>A. cabirinus</i>	36	68				Ethiopia Omo valley	MATTHEY (1968)
<i>A. cabirinus</i>	36	68	A	A		Egypt	TOBGY et al. (1972)
<i>A. cabirinus</i>	36	68	A	A		Egypt	DE-HONDT et al. (1977)
<i>A. cabirinus</i>	36 37	68 68	A A	A A		Sinai Sinai	WAHRMAN & GOTTEIN (1972)
<i>A. cabirinus</i>	38	68	A	A		Israel	WAHRMAN & ZAHAVI (1953)
<i>A. cabirinus</i>	36	68	A	A	C- C-	Ethiopia Rift Valley: a) Konso area b) Arba Minch c) Awash National Park d) 70 km to the North from Awash Nat. Park	Our data
<i>A. aurensis</i> <i>A. cabirinus</i>	42	70	A	M		Niger	TRANIER (1975)
<i>A. c. nesiotis</i>	38	68	A	A		Cyprus	MATTHEY (1968)
<i>A. seurati</i>	38	68				Southern Algeria	MATTHEY & BACCAR (1967)
<i>A. dimidiatus</i>	38	70	A	A +		Saudi Arabia	AL-SALEH (1988)
<i>A. minous</i>	38 40	66 68				Crete	MATTHEY (1963)
<i>A. chudeau</i>	40	70	SM	A		Morocco	BENAZZOU (1983)
<i>A. ignitus</i>	50	66				Kenya	MATTHEY (1956)
<i>A. wilsoni</i>	60	76				Ethiopia Omo valley	MATTHEY (1968)
<i>A. selousi</i>	60	70	giant SM	A		South Africa Malawi	MATTHEY (1965)
<i>A. spinosissimus</i>	60	72	SM	SM		South Africa, Transvaal province	DIPPENAAR & RAUTENBACH (1986)

(continued)

Table 1 (continued)

Species and forms <i>Acomys</i>	2n	NF	X-morphology	Y-morphology	Banding	Country locality	Authors
1	2	3	4	5	6	7	8
<i>A. subspinosus</i>	64	72	giant SM	A		South Africa	MATTHEY (1965)
<i>A. subspinosus</i>	64	74	giant SM	SM		South Africa, Cape province	DIPPENAAR & RAUTENBACH (1986)
<i>A. russatus</i> =	66	66	A	A		Israel	WAHRMAN & ZAHAVI (1953)
<i>A. levesi</i>	66	66	A	A		Egypt Jordan	QUMSEYEH et al. (1986)
<i>Acomys</i> sp.	68	68	A	A		Burkina Faso	GAUTIN et al. (1986)
<i>Acomys</i> sp.	68	68	A	A	C-	Ethiopia Koka area	Our data

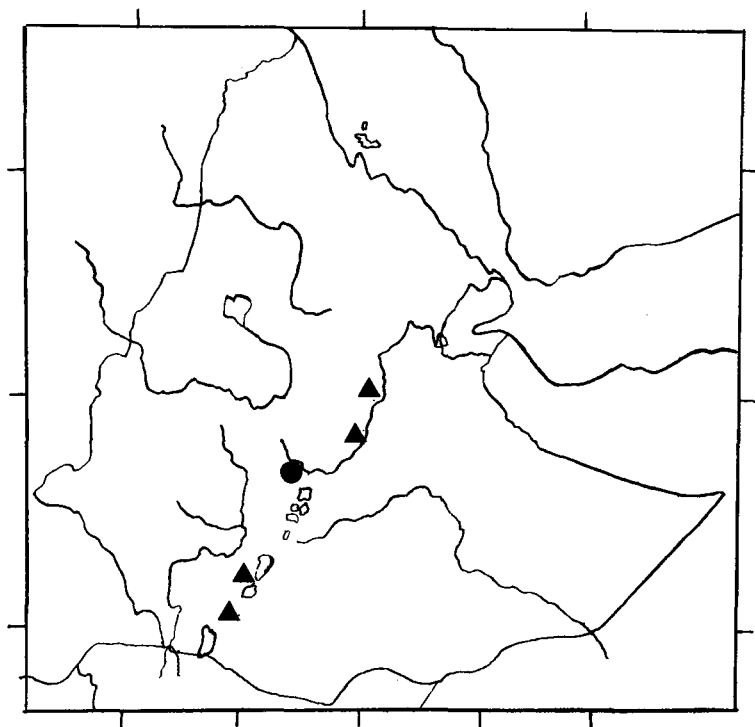


Fig. 1. — The karyological findings from the studies of *Acomys* in Ethiopia: filled triangle, 36-chromosomal *A. cabirinus*; filled circle, 68-chromosomal *Acomys* sp.

Park), 3 specimens; the valley of Awash river 70 km to the North from Awash National Park, 2 specimens (Fig. 1).

Somatic metaphases were prepared from bone marrow by the usual air-drying technique according to FORD & HAMERTON (1956), or by using short-termed tissue culture from dead animals (KOZLOVSKY 1974).

Slides were stained with 4% Giemsa in phosphate buffer pH 7. C-banding was obtained according to SUMNER (1972).

RESULTS

The karyotypes of the specimens of *Acomys* from Gamo-Gofa province (Konso area, Arba Minch) and from the middle part of Awash valley (Awash National Park, 70 km to the North from Awash National Park) belong to *A. cabirinus* (Fig. 1). The karyotypes are composed of 36 chromosomes comprising mainly harmed autosomes and acrocentric heterochromosomes ($NF=68$). The X-chromosome is the largest acrocentric and the Y-chromosome is the smallest one (Fig. 2A).

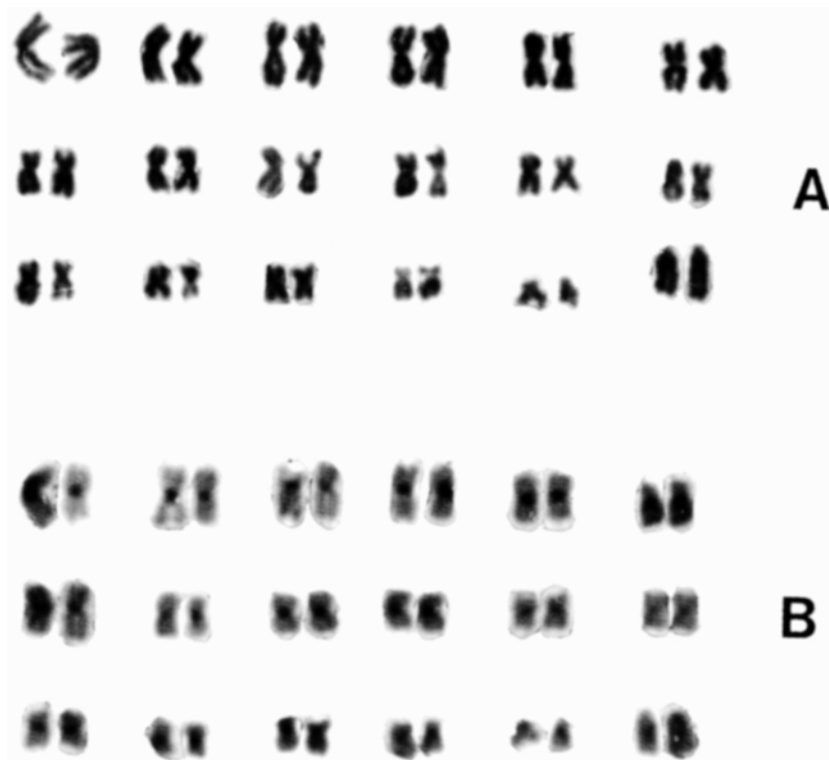


Fig. 2. — Karyotype of *A. cabirinus* from the Rift Valley of Ethiopia (Konso area): A, Giemsa staining, female; B, C-banding, female.

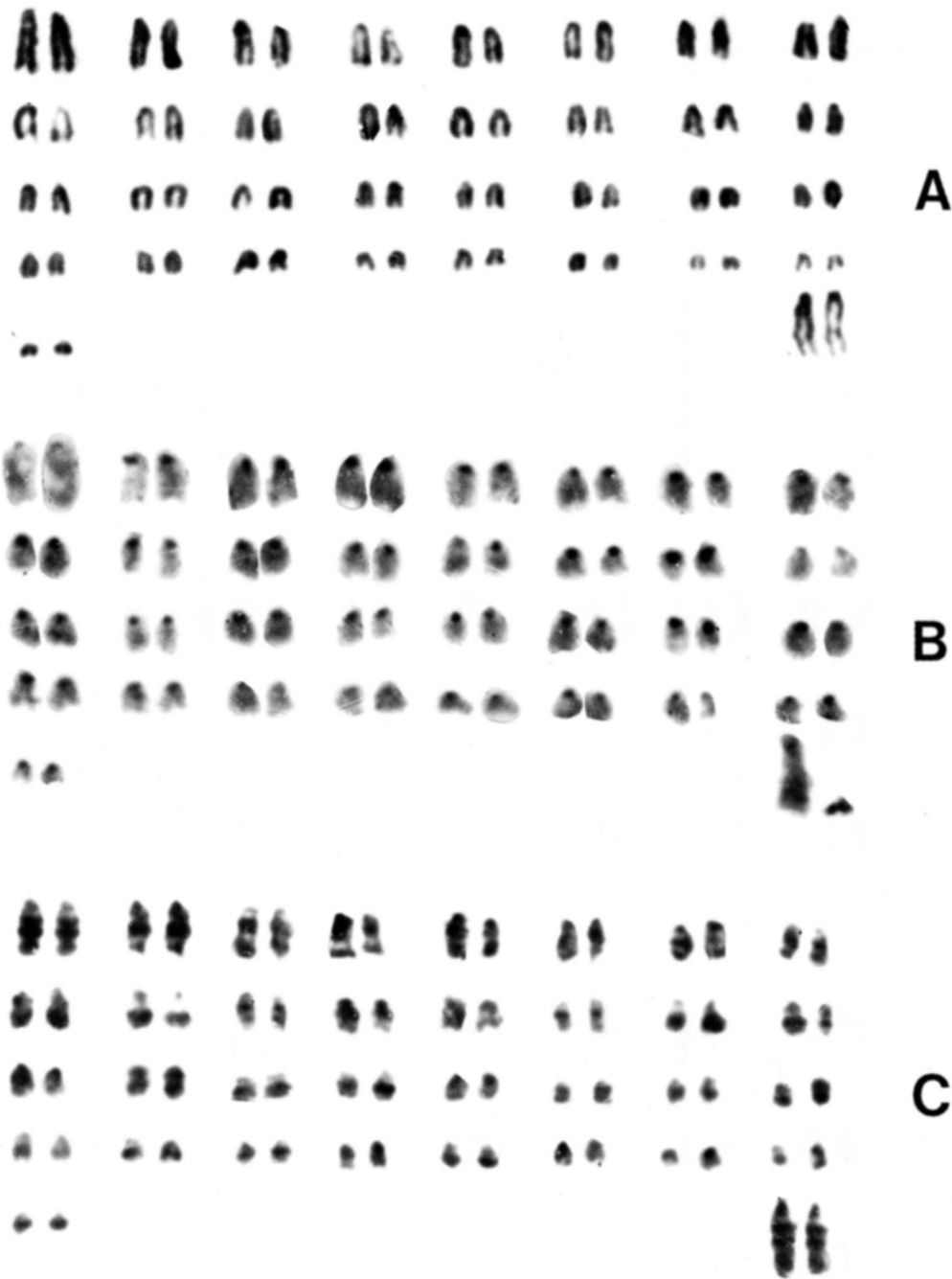


Fig. 3. — Karyotype of *Acomys* sp. from the Koka area in Ethiopia: A, Giemsa staining, female; B, C-banding, male; C, C-banding, female.

C-banding showed that almost all autosomes have small pericentromeric blocks of heterochromatin of different sizes and density. The X-chromosome was intensively stained in the pericentromeric regions and in the middle part of euchromatic arms. The Y-chromosome is highly heterochromatinized (Fig. 2B).

The karyotype of *Acomys* from the Koka area (upper part of Awash valley) (Fig. 1) belongs to the new form *Acomys* sp. This karyotype is composed of 68 acrocentric chromosomes. The largest acrocentric is the X- and the smallest is the Y-chromosome (Fig. 3A). All the autosomes and the X-chromosome showed blocks of pericentromeric heterochromatin with C-banding. The Y-chromosome is almost fully heterochromatinized (Fig. 3B).

DISCUSSION

Along the Ethiopian Rift Valley we discovered two karyotypic forms of *Acomys*: 36-chromosomal *A. cahirinus* and 68-chromosomal *Acomys* sp.

Previously, our knowledge of the karyology of the genus *Acomys* from Ethiopia was based upon the observations of MATTHEY (1968) on one specimen of *A. percivali* (= *A. cahirinus*) and one specimen of *A. wilsoni* from the southern part of the country (Omo valley). It was shown that *A. cahirinus* from the Omo valley had 36 chromosomes while that *A. wilsoni* had 60 chromosomes. In Ethiopia the distribution of *A. wilsoni* is limited to the southern part of the country, where its range overlaps with that of *A. cahirinus*. But *A. cahirinus* has wider distribution (YALDEN et al. 1976).

This is the first time that a 68-chromosome karyotype of *Acomys* sp. has been recorded in Ethiopia. Earlier, a diploid number of chromosomes $2n = 68$ was reported in West Africa (Burkina Faso) (GAUTIN et al. 1985).

The range of the 68-chromosomal *Acomys* sp. and its biotopic distribution needs further work. The specimens of *Acomys* sp. from the Koka area of Ethiopia were found in a semiarid rocky region in Acacia scrub at about 1500 m altitude. The specimens of *Acomys* sp. were collected in association with *Praomys fumatus*, *Tatera robusta*, *Arvicanthis dembeensis* and with rats from the complex *Mastomys natalensis*. But it is known that *A. cahirinus* in Ethiopia is characteristic of a similar biotopic distribution in arid desert and semidesert regions from sea level to 1500 m and rarely from altitudes greater than 2000 m (YALDEN et al. 1976). It is possible that *Acomys* sp. prefers semiarid regions. But additional information about the distribution of the 68-chromosomal form of *Acomys* sp. in Ethiopia for comparison with that of *A. cahirinus* and *A. wilsoni* is needed.

At present, it is not known if the range of *A. cahirinus* overlaps with that of *Acomys* sp. But it is clear that *Acomys* sp. and *A. cahirinus* are separate species. *A. cahirinus* with 36 chromosomes and *Acomys* sp. with 68 chromosomes represent the extreme variants of karyological differentiation in the genus *Acomys* (see Table 1). Indeed *Acomys* sp. ($2n = 68$) is characteristic of the highest diploid number of chromosomes in the genus *Acomys* with all acrocentric elements in its karyotype. *A. cahirinus* has the lowest number of chromosomes among *Acomys* that have been studied karyologically and has mainly biarmed elements in its chromosomal set. These two forms differ from each other by a minimum number of 16 chromosomal rearrangements, possibly Robertsonian or tandemic type translocations. At the same time we can suggest that heterochromatin variations did not make a significant

contribution to the separation of these two forms (pericentromeric autosomal and X-chromosomal heterochromatic blocks of different size and intensity were discovered in both karyotypes *Acomys* sp. and *A. cabirinus*).

Acomys sp. with 68 chromosomes is close to 36-chromosomal *A. cabirinus* in its morphological features such as the colouring of the fur (predominantly pale grey) and the dimensions of the body (the relative length of the tail is 80-105% of the body length). On the other hand, a chromosomal set with 68 chromosomes (*Acomys* sp.) is close to the species which possess a high number of chromosomes: *A. russatus* (= *A. levesi*) ($2n = 66$, WAHRMAN & ZAHAVI 1953) and *A. subspinosus* ($2n = 64$, NF = 66, MATTHEY 1965, DIPPENAR & RAUTENBACH 1986) (Table 1) but which belong to another group according to Petter's morphological classification. So there is disagreement between the karyological and the traditional approaches and a further revision of the genus *Acomys* is needed on the basis of all the data.

It is known that the evolutionary centre for the origin of spiny mice as determined on taxonomical grounds (ELLERMAN & MORRISON-SCOTT 1951) is the Ethiopian Region. Dispersion from this centre took place during the Plio-Pleistocene to different rocky and sometimes sandy habitats of Africa and during Pliocene to the same habitats of South-West Asia. By taking into account the course of chromosomal evolution in the genus *Acomys*, which is generally considered as an example of evolution by fusion (MATTHEY 1968, AL-SALEH 1988), it is possible to suggest that the 68-chromosomal *Acomys* sp. distinguished in Ethiopia is the form similar to the ancestor of this genus. This ancestor presumably radiated into a number of related species of which we can assume that the 36-chromosomal *A. cabirinus* is the most advanced. However, as long as the hypothesis of chromosomal dissociations or fission cannot be excluded. This last assumption remains questionable.

Further studies including multivariate chromosomal, biochemical, and molecular investigations are needed before definite conclusions can be drawn.

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